

v1.5 Reaxys crawler Guideline

1. Reaxys account registration

Address: <https://www.reaxys.com/#/institution>

When registering for the first time, the website will require the use of an institutional email for confirmation, so it is necessary to log in to the email during registration to complete the confirmation. After confirmation, registration is completed.

2. Compiler and environment configuration

2.1 Compiler

PyCharm 2022.3 (Community Edition) as the Compiler. Runtime version: 17.0.5+1-b653.14 amd64.

2.2 python environment

Table 1. Version of Python packages.

Name	Version
python	3.9.18
beautifulsoup4	4.12.2
pandas	1.5.3
selenium	3.141.0
lxml	4.9.2
openpyxl	3.0.10
pyperclip	1.8.2
urllib3	1.26.15

2.3 Firefox Browser Driver

This program uses Firefox as a simulated browser. First, you need to download the Firefox browser locally (do not download a too new version, otherwise there will be no corresponding driver).

Open Firefox to view the current version of the Firefox browser. exist <https://github.com/mozilla/geckodriver/releases> Find the corresponding version of Firefox on the website and download its driver. Put the drive into the Reaxys crawler

folder and replace geckodriver.exe in it.

3. Folder Introduction

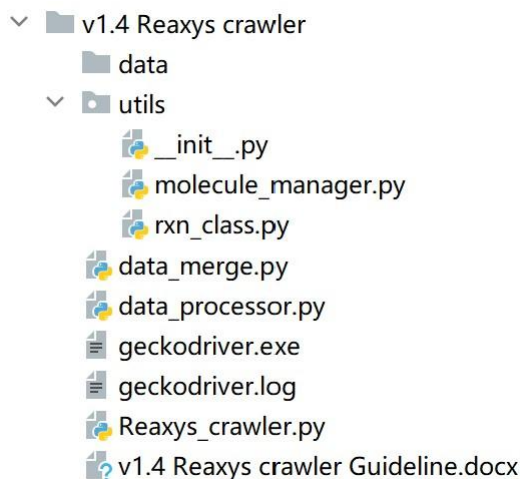


Figure 1. List of documents.

3.1 Three main programs

Reaxys_crawler.py: Simulate browser behavior, crawl reaction data from Reaxys website, and generate raw data.xlsx.

Data_processor.py: Preprocess raw data.xlsx, rename some compounds from IUPAC to SMILES, and process the data in the yield column.

Data_merge: Merge multiple files that store reaction data.

3.2 Data folder

Raw data.xlsx: Reaxys_crawler.py crawls websites and obtains data

Processed data.xlsx: The preprocessed data in raw data.xlsx is processed using data_processor.py

Merged data.xlsx: data_merge.py merges reaction data from multiple Excel folders

3.3 Utilis folder

Rxn_class.py: The RXN class is used to store reaction information; Convert the dataframe type that stores reaction data to a list type function, where reactions are

stored in the RXN class; Convert RXN class list to dataframe function.

Molecule_manager.py: A database tool that determines whether a molecule's IUPAC has been queried for its SMILES and records the IUPAC and SMILES of the queried molecule.

3.4 geckodriver.exe

Geckodriver.exe: Firefox browser driver.

4. Instructions for usage steps

4.1 Preparation Stage

Drag the folder to pycharm and open it. After configuring the environment, be sure to change the main in the upper right corner to Current File before proceeding with further operations.

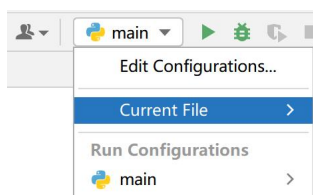


Figure 2. The place for “Current File” bottom.

4.2 Crawling of the html file of Reaxys

Open Reaxys_crawler.py and run it. Follow the prompts in PyCharm to enter the student ID, password, content to be searched, loading time for each webpage, and the number of pages to be crawled in sequence. The program will generate raw data.xlsx in the data folder.

Note:

(1) This program can only process data from Reactions search results. Please ensure that this column is included in the search results.

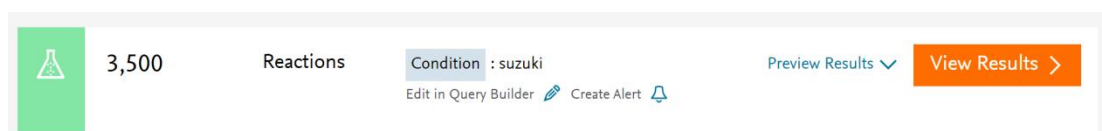


Figure 3. Searching of “Suzuki” as key word in Reaxys.

(2) Some compounds do not contain the SMIELS formula, and the program will skip this page directly and crawl to the next page.

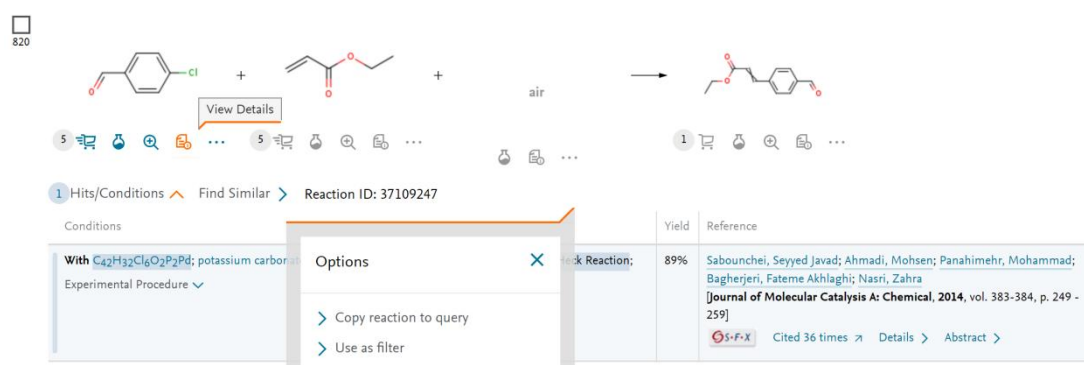


Figure 4. A case that the compound has no SMILES.

(3) Before entering the number of pages to be crawled backwards, you can find "Go to page" in the search bar of the simulated browser. After reaching the desired page, enter the number of pages to be crawled backwards in the Python program.



Figure 5. The “Go to page” bottom.

(4) Do not interfere with or close the simulated browser during the crawling process, otherwise it will cause crawling failure and prevent the generation of raw data.xlsx.

(5) The browser will automatically close after the program ends, and if it is closed halfway, raw data.xlsx cannot be generated.

(6) If the crawler reports an error while running, it is likely due to slow internet speed.

4.3 Data Preprocessing

Open `data_processor.py` and run it to get `processed data.xlsx` in the data folder. Data preprocessing is divided into two processes, namely the conversion of chemical substances from IUPAC to SMI format and the cleaning of yield column data.

The former is based on PubChem, address <https://pubchem.ncbi.nlm.nih.gov>. When there are no search results, the molecule will still follow its original naming format. At the same time, we have added a search mode using cactus, address: <http://cactus.nci.nih.gov>. This mode requires the use of VPN. If SMIELS cannot be found in this search mode, PubChem will be used for further search.

Due to some SMIELS copying failures (caused by the website), users need to search for or delete undefined data in the products and reactants through PubChem.

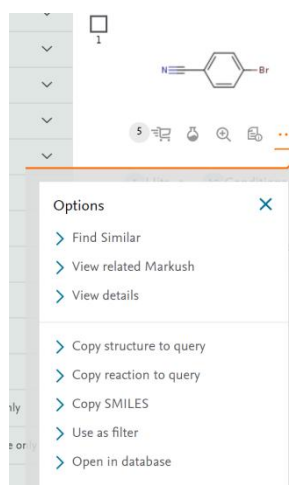


Figure 6. How to find “Copy SMILES”.

4.4 Data Merge

If you need to merge data using `data_merge.py`, you need to modify the code here yourself.

```
# 1. 打开文件夹
data = list()
data.append(pd.read_excel("./data/Heck_raw data 1-54.xlsx"))
data.append(pd.read_excel("./data/Heck_raw data 56-76.xlsx"))
data.append(pd.read_excel("./data/Heck_raw data 78-103.xlsx"))
data.append(pd.read_excel("./data/Heck_raw data 105-120.xlsx"))
data.append(pd.read_excel("./data/Heck_raw data 122-150.xlsx"))
data.append(pd.read_excel("./data/Heck_raw data 152-156.xlsx"))
data.append(pd.read_excel("./data/Heck_raw data 158-209.xlsx"))
```

Figure 7. Example for data merging.

5. Update logs

V 1.0 Implement Reaxys website crawling with v 1.0, export compound data named IUPAC, and convert IUPAC naming to SMILES expression.

V 1.1 Crawled out the reactants and products in the form of SMILES, and added inquiries about the number of pages crawled.

V 1.2 Updated bugs from the previous version; Fixed the issue of reporting an error when there is no Copy Smiles option in Reaxys; The issue of incorrect temperature units in the modified data table; The search engine for compound naming has switched to PubChem and removed the chk mode.

V 1.3 Optimized the operation flow of the program when there is no “Copy SMILES” option in Reaxys; Added crawling progress visualization; Added a data processing mode that uses both cactus and PubChem simultaneously.

V 1.4 Save data every 5 pages crawled, optimizing crawler stability; Changing the driver to Firefox avoids the situation where Chrome updates frequently and lacks the corresponding driver.

V 1.5 Translate the document to English Version.

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